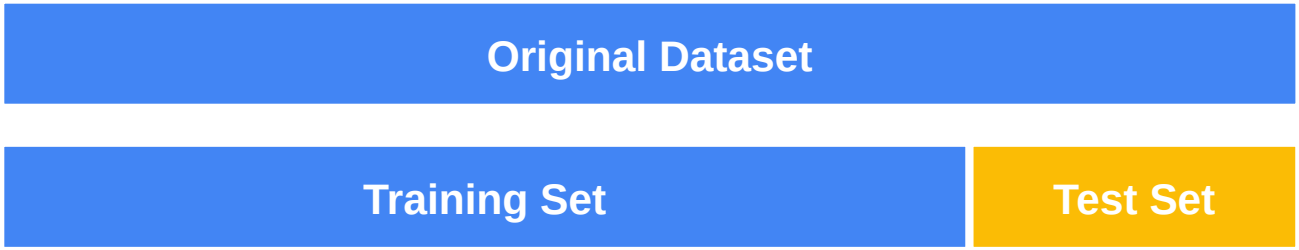
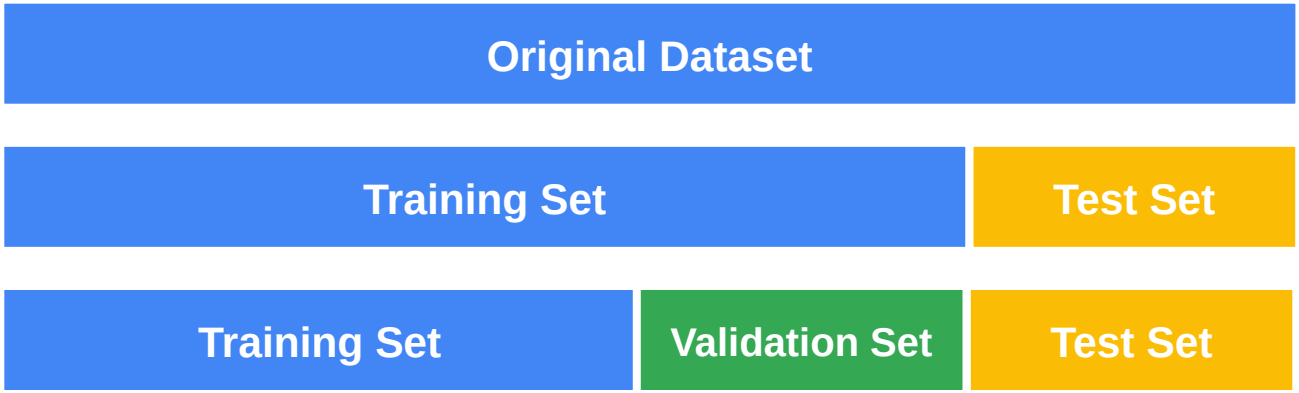
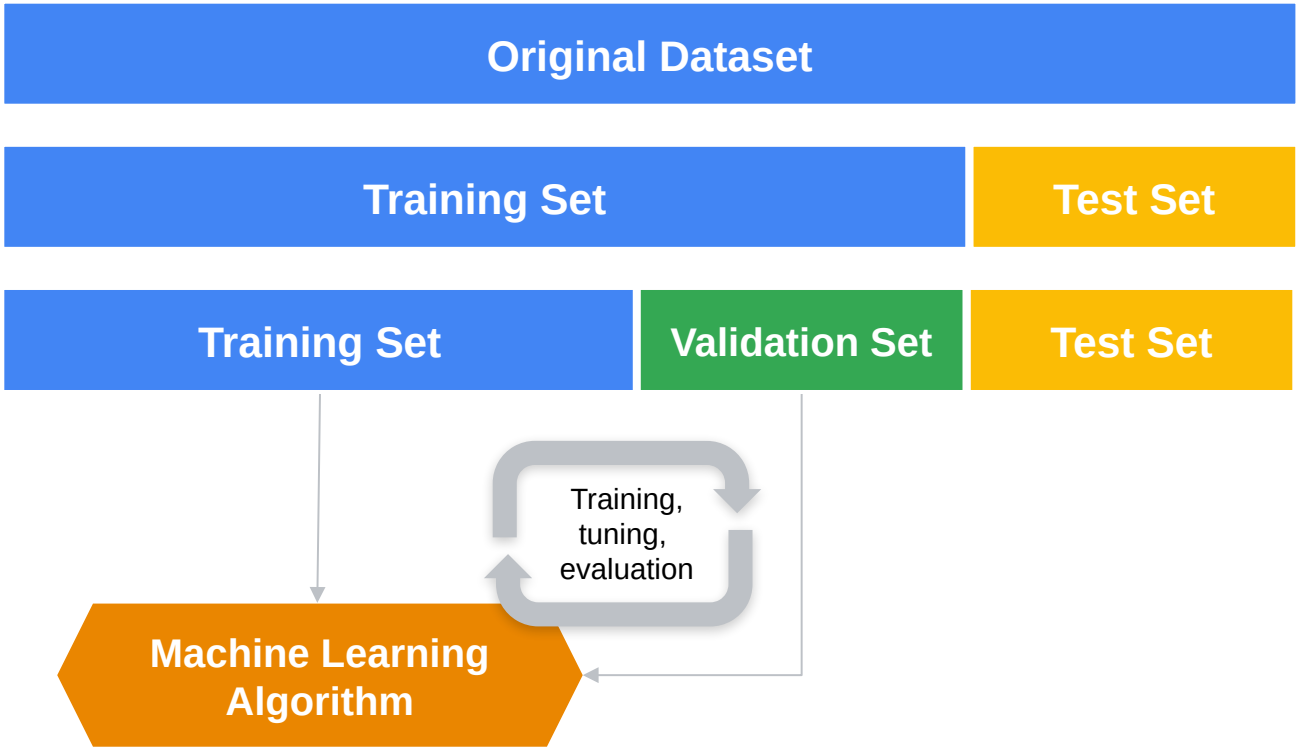
What is Data Engineering?

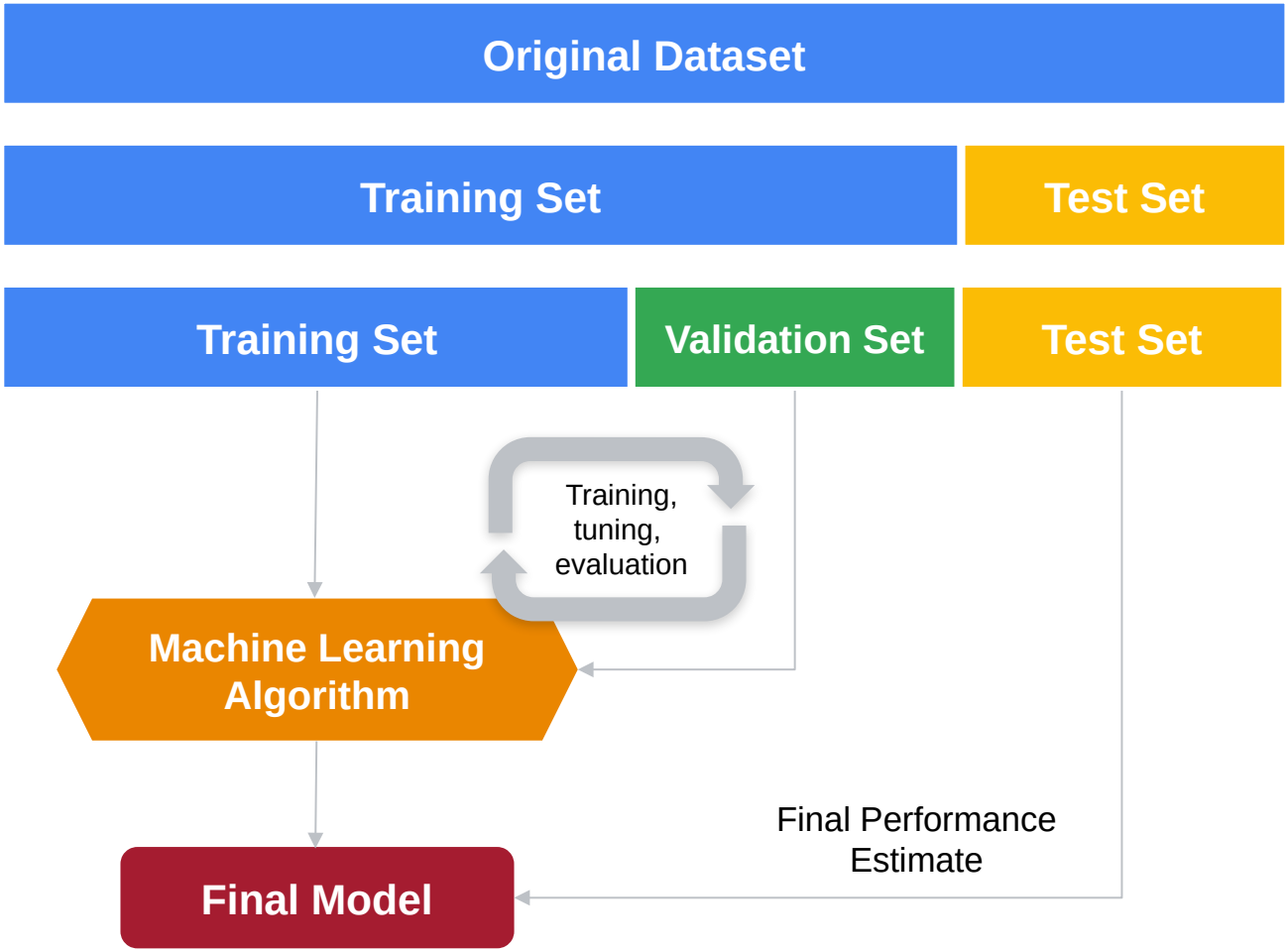
A supervised AI is trained on a corpus of training data.

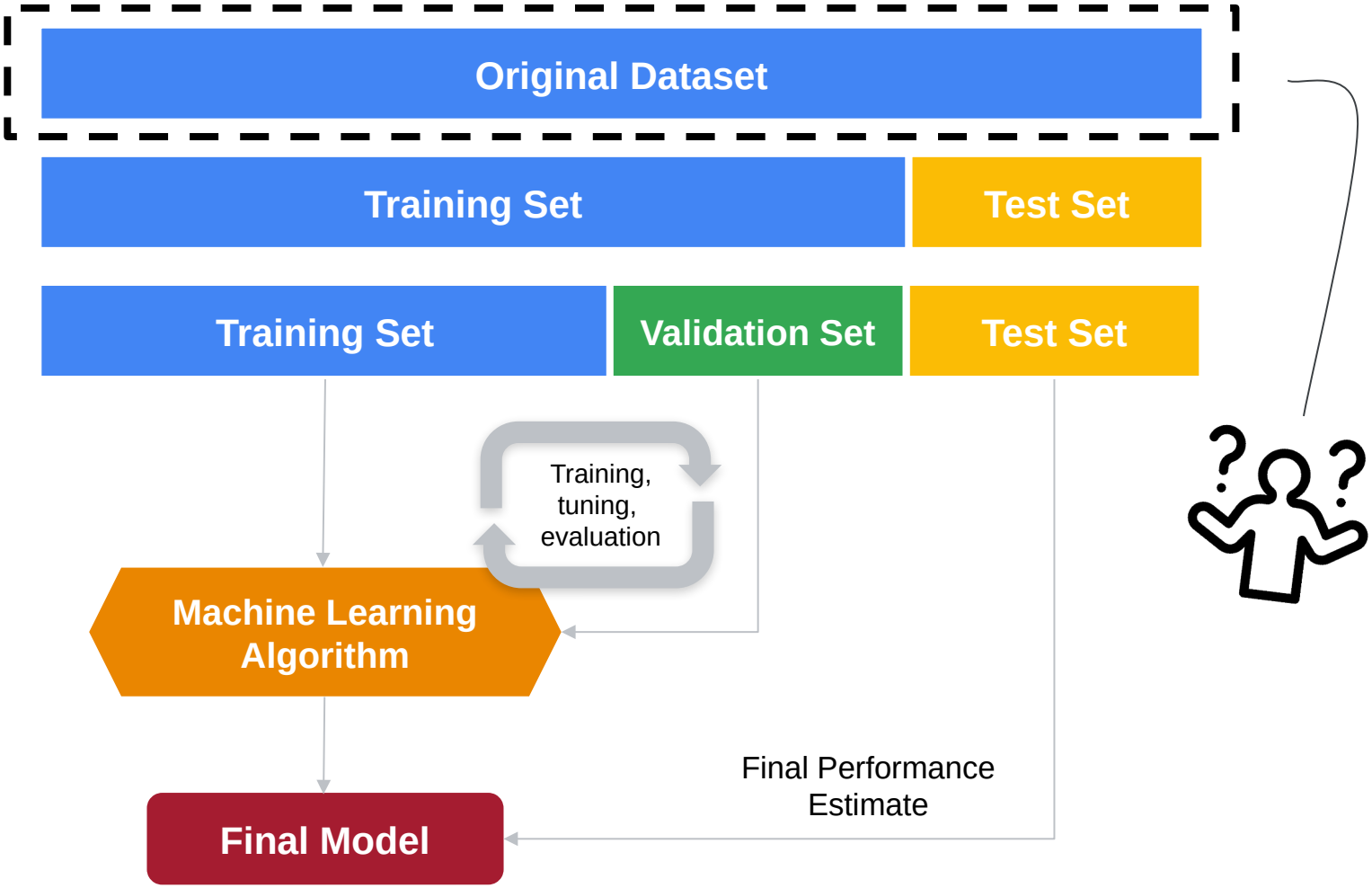
Original Dataset


Original Dataset
Training Set
Test Set


Original Dataset
Training Set
Test Set
Training Set
Validation Set
Test Set


Original Dataset
Training Set
Test Set
Training Set
Validation Set
Test Set
Training, tuning, evaluation
Machine Learning Algorithm


Original Dataset
Training Set
Test Set
Training Set
Validation Set
Test Set
Training, tuning, evaluation
Machine Learning Algorithm
Final Performance Estimate
Final Model


Original Dataset
Training Set
Test Set
Training Set
Validation Set
Test Set
Training, tuning, evaluation
Machine Learning Algorithm
Final Model
Final Performance Estimate

Data Engineering is all about **datasets**

Good Data is Necessary for Accuracy

What **problem** are you trying to ***solve***?

Good Data is Necessary for Accuracy

What **problem** are you trying to ***solve***?

- **Your data must contain useful features**

Good Data is Necessary for Accuracy

What **problem** are you trying to ***solve***?

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- **Can a human (expert) distinguish between examples of each class?**

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- **How will you measure performance?**

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Both ***quantity*** and ***quality*** will influence your model's performance

Good Data is Necessary for Accuracy

What **problem** are you trying to ***solve***?	• Your data must contain useful features • Can a human (expert) distinguish between examples of each class? • How will you measure performance?
Both ***quantity*** and ***quality*** will influence your model's performance	• **Wide distribution of training examples**

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- Wide distribution of training examples
- **Accurate labels**

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What **problem** are you trying to ***solve***?

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Both ***quantity*** and ***quality*** will influence your model's performance

- Wide distribution of training examples
- Accurate labels
- **Sufficient class balance**

Data Engineering

Requirements

- Problem definition
- Machine & human usable format
- **Permissions & rights**

Data **isn’t free** to use

Where does your data **originate**?

- Open?
- Copyrighted?
- Licensed?
- Product users?



What’s Yours and What’s **Not** Yours



Licenses







Data Engineering

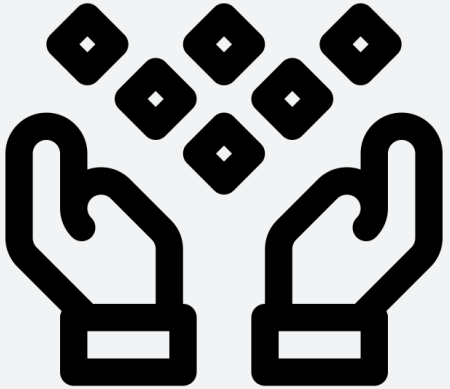
Requirements → **Gathering**

- Problem definition
- Permissions & rights
- Machine & human usable format

- People
- Collection
- Labeling
- **Data sources**

Data **sources**

- Sensors
- Crowdsourcing
- Product users
- Paid contributors



Data Engineering



- Problem definition
- Permissions & rights
- Machine & human usable format

- Data sources
- People
- Collection
- Labeling

- Processing
- Augmentation
- **Validation**

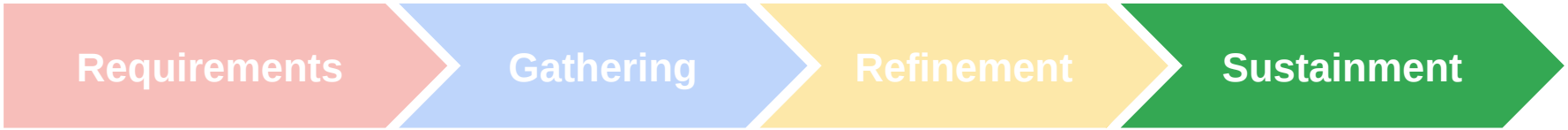
Some data is ***unusable***

How will you **verify** the data you collected?

- **Manually** (time, cost)
- **Automation**
- Domain expertise
    - disputes / disagreements



Data Engineering



- Problem definition
- Permissions & rights
- Machine & human usable format

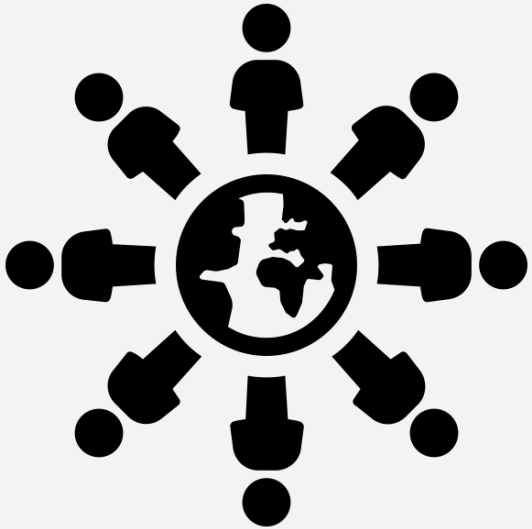
- Data sources
- People
- Collection
- Labeling

- Processing
- Validation
- Augmentation

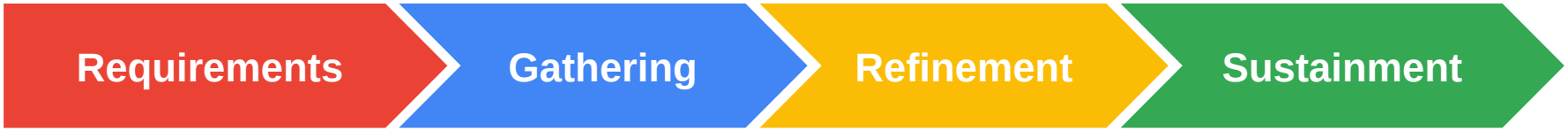
- Storage
- Security
- Errors
- **Versioning**

Your dataset will ***evolve***

- Missing **demographics**?
- **Expanding** your user-base?



Data Engineering



- Problem definition
- Permissions & rights
- Machine & human usable format

- Data sources
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- Storage
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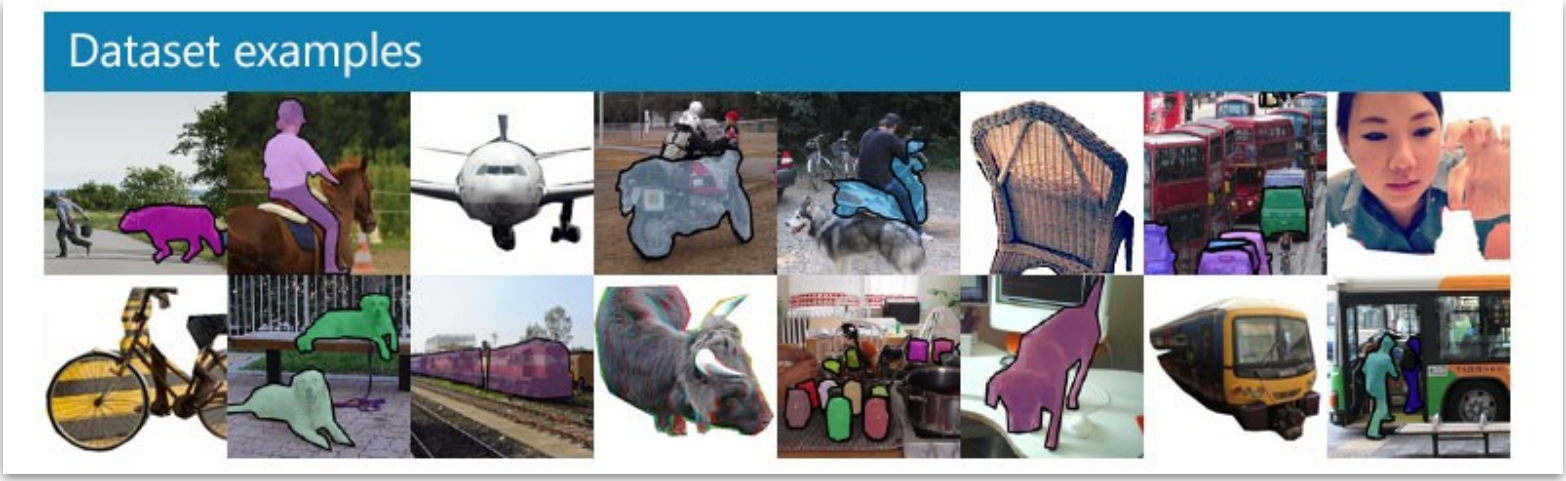
Datasets require **significant effort**

Classifying Images



Detecting Objects

- Common Objects in Context (**COCO**)—**2.5M+** segmented images



Datasets require ***significant effort***

- **Waymo**—**1,950** 20-second driving segments (cameras, LIDAR, labels)
- **KITTI 360**—**73KM+** of annotated driving data



Datasets require ***significant effort***

These **massive** machine learning datasets are ***constructed by hand***

- **Common Voice**—**5000+** hours of spoken audio
- Common Objects in Context (**COCO**)—**2.5M+** labeled images
- **ImageNet**—**4M+** labeled images
- **Waymo—1,950** 20-second driving segments
- **KITTI 360**—**73KM+** of annotated driving data

Data Engineering: *How to build your own dataset?*

How do you build your own datasets for ***TinyML***?